

Movie Ticket Booking System

William | Individual specification assignment | September 14, 2026

This report follows the project from competitive research and participant interviews through prototype evaluation and specification improvements. Participant responses are preserved in the interview appendix; design conclusions are presented separately.

Research

Competitive analysis

AMC Theatres. Published seat-selection pages provide screening context and sold-out recovery. Its refund FAQ routes customers through account tickets or confirmation email, and distinguishes refunds from editing an existing purchase. Design implication: keep the selected screening visible and explain cancellation before confirmation. [AMC seat selection](#); [AMC refund FAQ](#).

Fandango. Its official app listing describes seat-map preview, reserved seating and mobile tickets. Its policy explains account/email routes to refund or exchange, with eligibility and fee restrictions. Design implication: make the reservation reference easy to retrieve and communicate after-purchase rules clearly. [Official app listing](#); [Ticket policy](#).

Atom Tickets. Official support describes a movie/showtime-to-seat-selection-to-checkout flow and ticket retrieval. Purchase terms distinguish refunds from account credit. Design implication: preserve seat choices and total price through review; avoid unnecessary concessions and credit features in the first version. [Atom support](#); [Purchase terms](#).

Evidence boundary

These are public-documentation comparisons, not completed purchases or timed competitor usability tests. Some live pages restricted automated access; official indexed text was used where available. Recurring features support the proposed scope, but do not establish customer demand. The copied repository contains only a summary, so the complete business case still needs reconciliation.

Research

Technical feasibility check

Reliable seat allocation. PostgreSQL supports transactional row locks. A future app can atomically claim selected inventory rows, store a hold expiry and recheck it at confirmation. Database transactions should be short; the five-minute customer hold is stored state, not a five-minute database lock. Concurrent booking and repeated-confirmation tests remain required. [PostgreSQL locking documentation](#).

Payments. Hosted checkout is a documented future integration option. This assignment uses simulated reservations, no card fields and no charges. Real payment reconciliation and refunds would require separate implementation and testing. [Stripe Checkout](#).

Accessibility and hosting. Keyboard operation, clear focus, labeled controls and non-color seat states are explicit requirements. The paper prototype does not demonstrate WCAG conformance. GitHub Pages can serve the specification/static materials, but cannot provide the authoritative application database or account server. [WCAG reference](#); [GitHub Pages overview](#).

User feedback

William discussed the mockups with two family participants, P1 and P2, and supplied their interview responses. This convenience sample offers formative feedback rather than broad market conclusions. The interviews covered current purchasing, frustration, hold duration, group size, cancellation and desired improvements.

P1: Reports sometimes needing help with phone booking or using the box office. Unexpected fees, unclear screen orientation and pressure from the five-minute hold reduce confidence. P1 wants large simple text, clear totals and direct cancellation with an explicit confirmation.

P2: Reports frequent app-based purchasing with friends and a preference for concise text, strong visual hierarchy and recognizable seat states. Five minutes generally feels sufficient, but expiry should preserve the movie/showtime. Cancellation actions should differ in labels, symbols and appearance.

Different needs: P1 finds six seats restrictive for larger outings; P2 generally accepts the limit. The first version retains six seats while recording larger-group support as a future consideration. The complete supplied answers appear in the Research interview appendix.

Prototype Evaluation

Method and coverage

Five paper screens represent browsing/account handoff, seats, review/recovery, confirmation/cancellation and administrator validation. All fixtures are fictional. Participants point to actions; the facilitator reveals the next state. This satisfies the paper-prototype format, but provides no evidence of working payments, persistence, concurrency or production accessibility.

Desk inspection observations and revisions

- 1. Selecting is not holding.** The highlighted seats could be interpreted as already reserved. The seat sheet explains that Continue checks availability and starts the hold. Revised R6 explicitly requires that explanation.
- 2. Editing a held selection.** The review sheet originally offered Edit seats without saying what happens to the hold. The revised sheet warns that editing releases it and availability may change. R9 and its acceptance criterion now check this warning.
- 3. Preview versus cancellation.** The combined paper sheet risks confusing the confirmed reservation with the cancellation preview. The revised panel says Preview and Not yet cancelled. R14 requires separate keep/confirm actions and an unchanged reservation before confirmation.

These are desk-review observations recorded during preparation, not statements from William, P1 or P2. The initial v0.1 specification is retained; v0.2 records revisions and the review disposition of every template section.

Participant evaluation and resulting improvements

- P1 - hold duration:** The five-minute rule felt rushed. The revised design offers a calm time display and one optional two-minute extension. Its effectiveness needs a future retest.
- P2 - recovery:** Expiry should retain the screening and identify seats needing replacement. R8 now makes this explicit.
- Both - cancellation:** Keep and confirm-cancellation actions need an unmistakable distinction. The revised sheet separates their position, symbols, labels and colors.

These are qualitative prototype-review reactions; no timed completion, actual error count or observed timeout event was supplied. The revised design has not been retested.

Prototype Evaluation

My Evaluation

My first concern was keeping this assignment separate from the team project. I wanted my own copy so I could keep moving without changing anything the team was working on. Having the separate repository handled that.

During the build, the GitHub change view threw me off. It didn't look like a normal editor, and the temporary editing label made me question how the work was being saved. Opening the actual file and the readable page made the difference clear. That reinforced something useful for this project: the information can be correct, but if the presentation is unfamiliar, people can still feel unsure about using it.

I also checked the specification in desktop Chrome. After distinguishing the readable HTML page from the required Markdown link, I opened the .md version and confirmed it was live.

For the research, I wanted my wife's and daughter's responses kept in their own words, with the design decisions explained separately. Their feedback gave the project a clear direction: show the total earlier, make cancellation unmistakable, and give people a little more flexibility when reviewing their seats. Those improvements are now represented in the design; another review would help confirm how well they work.

Prototype Evaluation

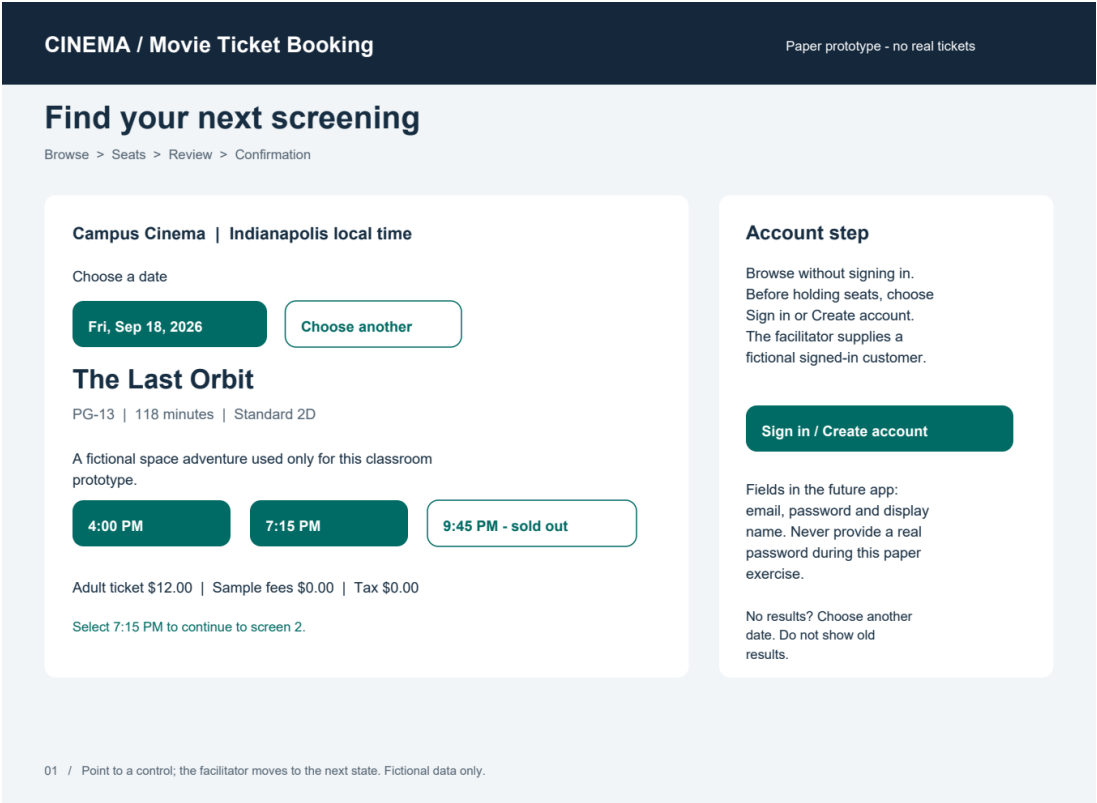


Figure 1. Browse a screening and explain the account handoff.

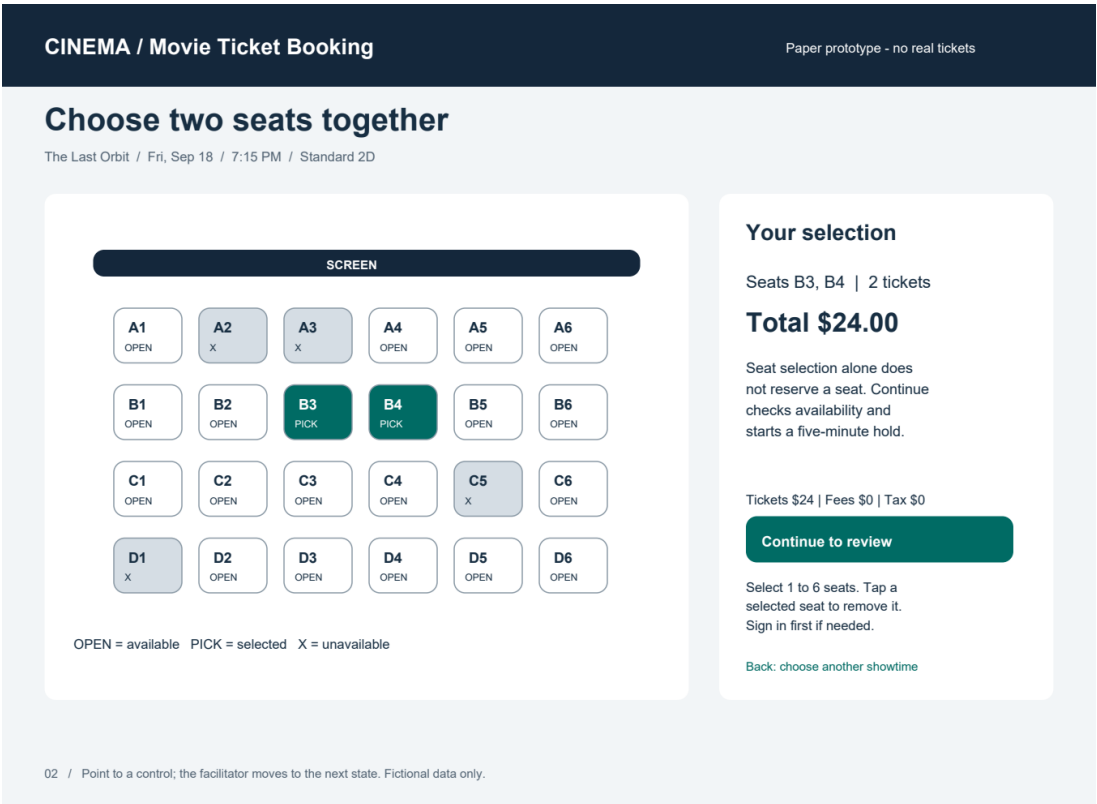


Figure 2. Seat identifiers, OPEN/PICK/X states and a visible total. The selected pair is a worked example.

Prototype Evaluation

CINEMA / Movie Ticket Booking

Paper prototype - no real tickets

Review before you confirm

Held seats > Check details and total > Confirm simulated booking

The Last Orbit

Campus Cinema - Auditorium 1

Fri, Sep 18, 2026 | 7:15 PM Indianapolis time

Standard 2D | Seats B3 and B4

2 adult tickets x \$12.00

\$24.00

Sample fees / sample tax

\$0.00

Total

\$24.00

Cancel the entire reservation before 7:15 PM on Sep 18. Seats are released. This prototype has no payment or refund.

Edit seats releases this hold; availability may change.

Confirm simulated booking

Edit seats

Time remaining: 01:00

A hold saves your seats briefly while you decide. Need more time?

Extend by 2 minutes

One extension, before time runs out.

If your hold expires

Keep your movie and showtime. Check the seat map again; seats may now be unavailable.

No confirmation? Check booking status before trying again.

Check booking status

03 / Point to a control; the facilitator moves to the next state. Fictional data only.

Figure 3. Review the total, hold deadline and cancellation rule; recover from uncertain results.

CINEMA / Movie Ticket Booking

Paper prototype - no real tickets

Your reservation is confirmed

Confirmation > My reservations > Cancellation preview

Confirmed - DEMO-1042

The Last Orbit | Campus Cinema, Auditorium 1 | Sep 18, 7:15 PM | Standard 2D

Seats B3, B4 | Total \$24.00

Find this reference again in My reservations. This is a demonstration, not a ticket for admission.

Open My reservations

Cancel reservation

Status remains Confirmed until you confirm cancel.

Preview: cancel DEMO-1042?

Not yet cancelled. Review before confirming. This cancels both B3 and B4 for The Last Orbit on Sep 18 at 7:15 PM. You cannot cancel only one seat.

No money was charged. Both seats will be released. Cancellation is allowed only before showtime.

< Keep reservation

X Confirm cancellation

After confirmation: Cancelled. Seats released.

At/after showtime: cancellation unavailable.

04 / Point to a control; the facilitator moves to the next state. Fictional data only.

Figure 4. Confirmation and cancellation preview are separate states shown together on a paper sheet.

MOVIE TICKET BOOKING / INDIVIDUAL ASSIGNMENT

6

Prototype Evaluation

CINEMA / Movie Ticket Booking

Paper prototype - no real tickets

Manage movies and showtimes

Administrator role only / supporting flow for the initial specification

Add a showtime

Movie

The Last Orbit (118 min)

Auditorium

Auditorium 1

Date / local time

Sep 18, 2026 - 7:15 PM

Format / price

Standard 2D - \$12.00

Validate and publish

Validation feedback

If another screening overlaps runtime plus 15-minute turnaround, explain the conflict and keep the entered fields.

If the showtime already has active reservations, reject changes to its time or auditorium.

Successful save: Published. The screening appears in the customer date list.

Customer role: access denied. Administrative controls are unavailable.

05 / Point to a control; the facilitator moves to the next state. Fictional data only.

Figure 5. Supporting administrator flow with overlap and booked-screening validation.

Specification

Version 0.3 incorporates the participant feedback and prototype inspection. It contains all ten template sections, five scenarios, 20 requirements and matching acceptance criteria. R6 strengthens early price itemization, R8 adds extension/recovery, R14 differentiates cancellation controls and R19 strengthens readable text and visual cues. Owner sign-off remains pending.

[Open readable specification on GitHub Pages](#)

[Required Markdown source URL](#)

[Open specification in the repository](#)

Browser verification: William confirmed the exact specification.md URL live in desktop Chrome and provided a screenshot showing version 0.3.

The required Markdown link was confirmed live by William in desktop Chrome. The report includes personal evaluation, participant responses and prototype images. The specification records remaining product questions and the need to reconcile the complete business case. The original team repository remains untouched.

Research

Interview responses - Participant 1

Responses supplied by William; wording retained as provided.

Context: Books weekend outings for her family; primarily uses her phone for calls, text, and basic browsing; easily stressed by rushed online transactions.

1. How do you usually book movie tickets today, and what frustrates you most about the process?

"I usually try to do it on my phone, but half the time I get overwhelmed and make my husband do it, or we just show up at the theater box office. The worst part is getting to the very end of checkout and suddenly seeing \$15 in mystery service fees tacked on. Also, picking seats is stressful—I can never tell where the screen actually is compared to the seats, and if I tap the wrong seat by accident, it freezes or loses my spot."

2. The system holds your seats for 5 minutes while you review and confirm. How does that feel to you?

"Five minutes feels like a fire drill! If I'm digging through my purse for my wallet, checking with my family in the other room about who wants which seat, or trying to read the fine print, five minutes goes by in a flash. A giant countdown timer ticking down makes me panic that I'm going to make a mistake."

3. What do you think about being able to book a maximum of 6 seats at once?

"Six is fine for just our family, but if it's a birthday party or we go with the neighbors, six isn't enough. We'd have to do two separate orders, and there's no guarantee we'd get seats next to each other on the second try."

4. How would you expect cancellation to work if plans change?

"Life happens—kids get sick or dinner runs late. Being able to cancel right from my account before the movie starts is great, but it needs to be completely obvious. Don't make me hunt through nested menus, and please make sure the button clearly says 'Confirm Cancellation' so I don't accidentally wipe out my tickets while just checking the policy."

5. What is the most important thing this website could do to make you feel comfortable using it?

"Make the text large and simple. Tell me the actual total price right up front instead of hiding fees, and if I make an error, tell me in plain English what went wrong instead of giving me a red box with tech jargon."

Research

Interview responses - Participant 2

Responses supplied by William; wording retained as provided.

Context: Uses Discord, gaming platforms, and mobile apps daily; navigates digital interfaces rapidly by relying on layout, icons, and visual hierarchy rather than reading dense text.

1. How do you usually book movie tickets today, and what frustrates you most about the process?

"Me and my friends usually buy tickets through apps on our phones right before we meet at the mall. The biggest pain is walls of text. When an app puts a giant block of paragraph text explaining theater rules or pricing breakdowns, the words just scramble together. I just want visual cues—big buttons, clear movie posters, and high contrast."

2. The system holds your seats for 5 minutes while you review and confirm. How does that feel to you?

"Five minutes is plenty of time for me to click through, but if the hold expires, please don't just dump me back to the homepage. If my hold runs out while I'm trying to read the confirmation page, keep the movie and showtime picked and just tell me visually: 'Seat B3 was lost, tap a new seat.'"

3. What do you think about being able to book a maximum of 6 seats at once?

"Six works for our group most of the time. If someone else comes, they just buy their own on their phone next to us. It's totally fine."

4. How would you expect cancellation to work if plans change?

"I like that you can cancel before showtime, but the screen design has to be foolproof. If the 'Keep Reservation' and 'Cancel Reservation' buttons are identical gray boxes, or if there's no color and icon difference, I might misread them and hit the wrong one. Use distinct shapes, symbols (like a checkmark vs. an X), and strong colors so I don't rely purely on reading the labels."

5. What is the most important thing this website could do to make you feel comfortable using it?

"Keep the seat map clean. Don't use faint pastel colors to show available versus taken seats—use bold shapes, distinct icons (like a filled square vs. an open circle), and clean fonts with good spacing. If I can understand the screen at a glance without having to decode small text, it's 10 times faster."